Disparities in Cancer Treatment and Patient Outcome in Rural Communities in the United States

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On my honor as a University Student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments

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Introduction

Cancer is the second largest leading cause of death in the United Sates, resulting in 619,876 deaths in 2024 in the US alone (Centers for Disease Control and Prevention, 2026). The National Cancer Institute (NCI) estimated that in 2025, over 2 million new cancer cases would occur in the US (National Cancer Institute, n.d.). Both the cancer case rate and the cancer death rate have decreased over time due to advancements in treatment and as causes for increased cancer risk are more widely communicated. Even within the past decade, improvements have been made regarding new, effective types of immunotherapies and the link between smoking and obesity to cancer has become clearer and more widely communicated (McDowell et al., 2019). Despite these improvements, the rate of cancer deaths in rural areas is higher than in urban areas, with a 5-year survival rate of 62% in rural areas and 67% in urban areas (Islami et al., 2023).

Worse patient survival outcome in rural areas can be indicative of deficiencies in the medical systems in those areas when compared to urban areas. These factors can include decreased presence of health care professionals, fewer available treatment options, lower screening rates, and economic barriers to treatment and screening. This paper will explore the severity and prevalence of these factors in rural communities and their effects on patient outcome. Additionally, I will present some of the proposed solutions to these factors and interpret the exact issue that is being targeted and the potential barriers to implementation or efficacy of the solutions.

Methodology

This paper will examine how socioeconomic and environmental factors impact rural patient cancer outcomes. The method for analysis is a qualitative analysis from existing literature, primarily pulled from Google Scholar and government and relevant organizations such as the Centers for Disease Control and Prevention (CDC) or the American Cancer Society (ACS). The literature review will first focus on the deficiencies regarding the quality and accessibility of health care access in rural areas. I will then address the impact that socioeconomic conditions have on cancer treatment and their impact on patient outcomes. Finally, I will explore some solutions that have been proposed and discuss the barriers that could reduce the impact of the solutions or prevent them from getting off the ground entirely.

Quality and Accessibility of Care

One driving component of inequity in cancer patient outcomes in rural areas is the lower quality of care and the inaccessibility of care. The deficiencies include a lack of healthcare professionals, not being within proximity to care, and limited participation in clinical trials.

Rural areas in the US have fewer health care professionals per capita than their urban counterparts, including for oncology. While approximately 15% of the US population resides in rural areas, only 5.6% of oncologists practice in rural areas (CDC, 2024). This results in a higher patient to medical professional ratio; 6.5 oncologists per 100,000 people in rural areas compared to 16.6 oncologists per 100,000 people in urban areas (Davenport, L. A., 2025). Additionally, rural hospitals also have higher patient-to-nurse ratios than urban hospitals, with rural hospitals averaging a ratio of 7.3 and urban hospitals averaging 4.8 (Smith et al., 2019). Cancer patients that are cared for by nurses with higher patient-to-nurse ratios tended to have higher 5-year mortality rates than those with lower patient-to-nurse ratios (Han et al., 2024).

The decreased presence of healthcare professionals in rural areas contributes to worse patient outcomes. Less oncologists dedicated to a population can result in longer waiting times to be seen by an oncologist for screening as well as less personalized care for a patient who has been diagnosed with cancer. Longer waiting times can result in cancer progression which results in higher 5-year mortality rates. A cancer patient with more progressed cancer will need more intense and frequent care. This additional care can also be pushed back due to fewer available oncologists. Additionally, the nurses that care for the patient may not be able to give adequate attention to those that need it as they often must spread their time among other patients more thinly.

Cancer screening and treatment are often limited by the patient's proximity to the nearest facility that can administer treatment or screen for cancer. A study conducted surveyed over 52,000 women with breast cancer over the course of 11 years found that women in rural areas traveled distances 3 times larger to receive adequate care than women in urban areas, with women in rural areas traveling an average of 40.8 miles (Longacre et al., 2019). Longer travel distances to treatment put a larger burden on rural patients and often reduces their options to travel to treatment sites, especially without a car. Public transit routes often do not have routes to take patients from sparsely populated areas to larger areas due to minimal demand and ridesharing options often become expensive and unreliable to ensure patients are able to arrive on time to their appointments. This inaccessibility to treatment sites often results in worse patient outcomes as many patients are forced to miss or reschedule appointments (Wercholuk et al., 2022). Missed and rescheduled appointments can result in delayed care or screening, worsening the patient’s outcome. The large travel distance may also push patients away from follow-up appointments and recurrent appointments for treatment. When a patient is unable or unwilling to travel large distances for treatment, they will stray from the treatment path set in place for them which may include several visits to a site within a short amount of time. Nonadherence to recommended treatment plans often results in worse patient outcomes including higher mortality rates. Inaccessibility to treatment sites is expected to worsen as rural hospitals continue to either reduce offered services or close all together. Over the span of two decades, there were 195 hospital closures in the US, a map of which can be seen below in Figure 1. This includes 110 complete closures and 85 converted closures.

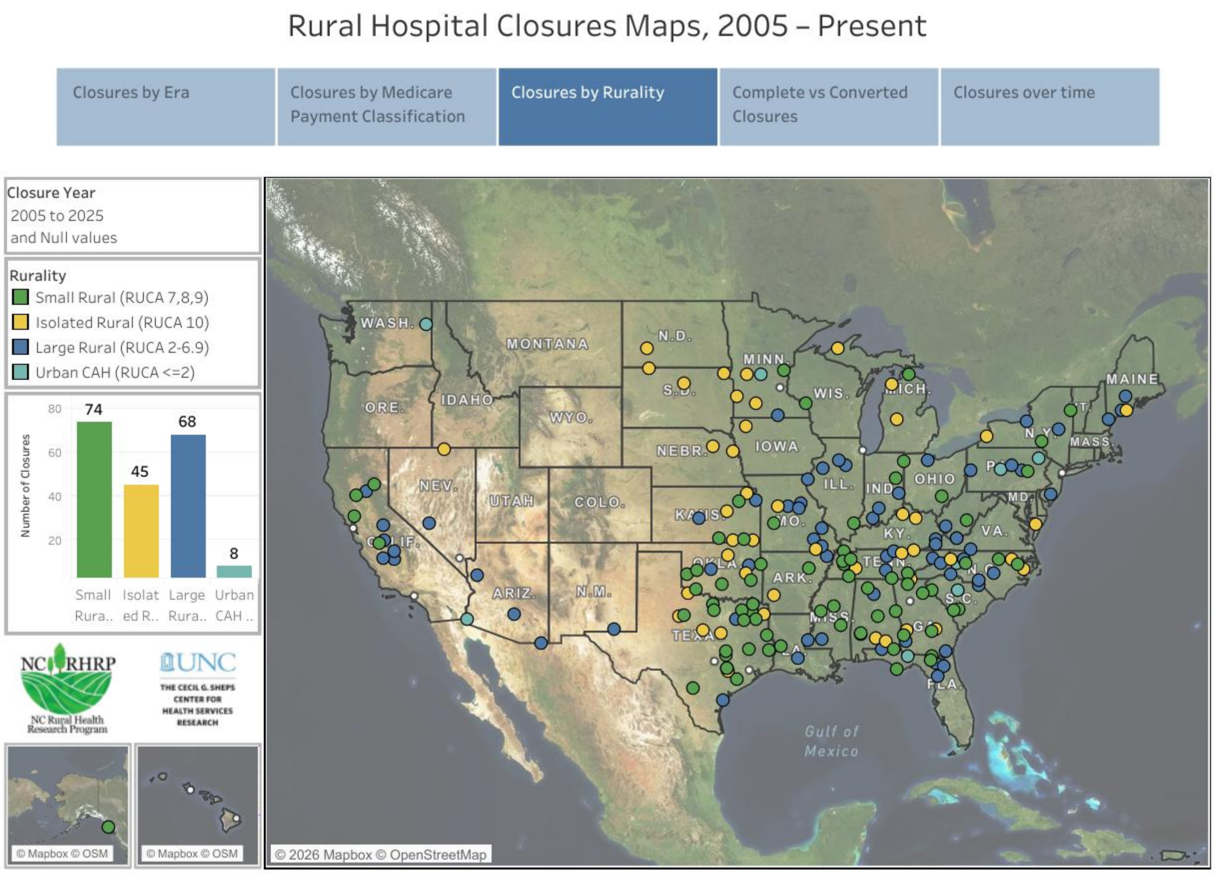

Figure 1: Map of hospital closures in the United States between 2005-2025 (University of North Carolina, 2025)

Figure 1 also includes a bar graph that shows the number of hospital closures by hospital location and size: small rural, isolated rural, large rural, and urban critical access hospitals (CAH). Of all hospitals closed over the last 20 years, 95.6% of them were rural hospitals. The continued closure of rural hospitals will ultimately increase the distance rural patients will need to travel, resulting in more people that may not adhere to their recommended treatment plan.

Clinical trials are imperative to cancer treatment as they allow new methods of treatment to be explored, many of which prove to be effective. However, many rural hospitals do not have a way for patients to participate in clinical trials at their location. This results in additional travel for the patients. One study conducted on 179 patients found that the average travel distance for the rural patients to clinical trials was 110 miles, an incredible burden to have to take on (Sabesan et al., 2010). Additionally, the same study found that rural patients were just as willing to participate in clinical trials but often found that both the large travel distance and the need to bring someone with them to take them back after treatment presented a much larger barrier to clinical trial participation. Despite the willingness to participate in these trials, most rural hospitals lack the resources to administer them. This can be due to a lack of funding, inadequate staffing, non-specialized equipment, and smaller groups of people to participate. Additionally, any clinical trial that is funded by the NCI must be administered at an NCI-designated cancer center or at National Institute of Health (NIH) Clinical Center, all of which are in large metropolitan areas. The technical requirements to run clinical trials and the distance from capable centers introduce a barrier that prevents rural patients from participating in potentially life-saving clinical trials.

Socioeconomic Impacts

The physical factors of cancer treatment are not the only barriers to improved patient outcomes for rural patients. Socioeconomic factors such as patient income relative to the cost of treatment, non-care related costs, and insurance coverage can also impact 5-year survival rates. In rural areas, 49.3% of cancer survivors reported economic hardships while only 38.7% of urban cancer survivors reported economic hardships, are indications that negative economic factors contribute to worse patient outcomes (Zahnd et al., 2019).

The average salary in rural areas is lower than that of urban areas, $45,917 and $61,717 respectively as per the 2020 US census (ACS, 2022). However, the cost of cancer care is generally independent of rurality. This results in care that is more expensive relative to the patient's income, making treatment a larger burden. Treatment costs that are too high may result in nonadherence to recommended treatment. Additionally, economic barriers can decrease the likelihood that a patient will have up to date cancer screening performed. A survey conducted in the state of New York examined how income impacted breast cancer screening among women. They found that women who made <182% of the 1-person federal poverty level were screening at a rate of only 68.4% compared to women with high incomes who had a screening rate of 82.5% (He et al., 2020). The decrease in cancer screening can result in cancer being caught at a later stage at which treatment is not only more expensive, but often more intensive and therefore expensive. The higher relative cost of cancer treatment for rural patients can result in fewer screenings and decreased ability to pay for life-saving treatment.

While the treatment itself can already prove to be a larger economic burden for rural patients than urban patients, rural patients often find there are additional costs associated with care. One such cost is the added cost of transportation. In the previous section, I highlighted that many rural hospitals do not have the necessary resources to treat patients, requiring them to travel great distances to receive adequate care. When assessing the willingness for clinical trial participation for rural cancer patients, many expressed concerns regarding the cost or travel. In the survey, 41.1% of rural participants expressed concern regarding the cost of travel while only 23.5% of regional participants expressed that same concern (Sabesan et al., 2010). Some of these costs can include gas, tolls, and the increased need for car maintenance. The price can grow for those without a car as well. Without public transit, a patient may need to utilize a rideshare service or find someone who is able to drive them to their appointments where trips are often over 50 miles one-way. This economic burden may prevent patients from adhering to their treatment plan and result in skipped or delayed treatments, both of which can result in higher patient mortality rates.

Another non-care associated cost is the time lost from work, which often results in a decreased income. According to the ACS, 74% of patients who received care missed days they would have normally worked, with 39% of those who missed work reporting they lost 13 weeks or more of time at work. 32% of those who missed work for treatment reported they received no income for some or all the missed time. 16% of respondents either lost their job, quit, or had their hours reduced (ACS, 2025). A loss in income is already greatly impactful when receiving cancer treatment, but this is felt especially hard in rural areas where the average salary is lower. A reduction of income can push patients to forgo screening or delay or minimize treatment to reduce the financial burden.

Insurance coverage also influences the economic impact of cancer treatment as it may influence which treatments a person may use, whether they get screened, and often, if they peruse treatment at all. Approximately 14% of rural Americans are uninsured, 20.3% have some form of public insurance, and 57.9% are privately insured (ACS, 2022). For uninsured patients, they have to cover the entire cost of treatment or screening which can often be a financial burden that results in patients either forgoing treatment or screening or going into debt to receive treatment or get screened. Of the 57.9% of privately insured people, 43% of them are enrolled in a high-deductible health plan (HDHP). HDHPs require much higher amounts of out-of-pocket contributions to health care before the insurance plan covers anything. Many rural Americans chose HDHPs to minimize the recurrent necessary expense of health insurance. While HDHPs are the most cost-effective when a person is perfectly healthy, they result in much higher costs for the patient once they are no longer healthy. These high costs can result in patients choosing to forgo treatment or to choose less effective treatment that is available in-network.

Rural patients have also reported that provider networks in rural areas are more restrictive than those in urban areas (Feyman et al., 2024). Increased restrictiveness is a result of limited providers within a rural area, decreasing the number of resources that are covered under insurance. If a patient's treatment requires them to go to a specific clinic outside of the general area they live in for treatment, it is often out-of-network and will not be covered by insurance, increasing the financial burden on the patient. Absorbing the entire cost of treatment due to a restrictive insurance provider network may be something patients are unable to do. This may result in them choosing a different treatment path that may not work as well as the recommended out-of-network path. This can also result in lower patient survival rates.

Potential Solutions

As discussed in the previous two sections, there are many systemic issues that result in rural cancer patients experiencing worse patient outcomes. However, some solutions have been proposed to address the treatment quality, treatment availability, and socioeconomic barriers of cancer treatment. In this section, I will introduce some of the solutions and explain what they are attempting to address as well as any barriers to their effectiveness. I will be pulling the potential solutions from a paper that aims to distinguish where rural cancer care can be improved (Unger et al., 2025).

Unger seeks to address treatment quality and issues with regional care limitations by proposing an increased partnership between rural hospitals and larger networks. He described the approach as a hub-and-spoke model: there is one central hub for an area (the hub) which provides more intensive and complex care and that hub is connected to several smaller hospitals (the spokes) which will provide basic care and direct to the hub when they are unable to provide the necessary treatment. This model has advantages as it allows for care that is tailored to the area and community at each spoke and provides basic, more routine treatment within the local community. It is then able to easily connect patients to the larger hub when the spoke is unable to provide treatment which can make the process smoother. It can also have the added benefit of increasing the scope of hospitals that an insurance company may consider in-network as they work directly with one another to provide care to the patient. However, this model has some deficiencies. It does not eliminate the additional burden of travel for rural patients, as they will still be required to go to the hub for some treatment. The hub would likely be the hospital they would have traveled to prior to the hub-and-spoke model. The cost of travel as well as the time away from work for rural patients would still be higher than that of urban patients. This model would also require coordination between the hospitals which may incur costs for the hospitals which could be passed to a patient's bill.

Unger proposes that subsiding drug prices and promoting clinical trial usage in rural hospitals would allow for more people to access drugs that may not have been covered by their insurance.

Subsidizing drug prices would increase drug accessibility by reducing the price enough such that patients are able and willing to receive the best treatment for their needs. While this would increase accessibility by reducing prices, the implementation of this would require federal government involvement. Specifically, the 340B Drug Pricing Program would need to be amended to consider subsided drug pricing on a per-patient basis rather than the hospital. This would allow for uninsured or underinsured people to access the drugs for a reduced cost. However, this is a change that would require congressional action and would require the movement to garner support from both politicians and constituents.

Promoting clinical drug trials in rural hospitals would address some issues regarding the travel burden as well as reduce the likelihood the clinic would be out-of-network. It would entail that pharmaceutical companies which do most of the clinical trials in the US to bring clinical trials to rural hospitals or by reimbursing rural patients that need to travel to participate. Despite these advantages, clinical trial locations are chosen by pharmaceutical companies due to large population centers providing a large sample size with minimal environmental impacts of trial outcome. Additionally, they would have to fund these trials in the rural communities which include ensuring they have access to necessary equipment and that those administering the trial have proper training. At the end of the day, pharmaceutical companies are a business and will most likely make decisions that save them money.

To improve access to appointments, Unger recommends the usage of telehealth appointments to allow for patients to consult with oncologists without the added costs or waiting time associated with in person appointments. This solution has been implemented in many rural communities, and it has resulted in patients reported improved care quality. Both patients and oncologists have expressed that the consultations are satisfactory as they improve patient accessibility and help patients build a relationship with their care team while also reducing costs (Morris et al., 2021). This approach has strong positive implications for consultation and support, but it is unable to address the physical component of cancer care, which would need to be addressed through another model such as hub-and-spoke.

Conclusion

Cancer treatment in the rural United States is less robust than in densely populated urban areas. Throughout this paper, I have argued that the factors that result in lower 5-year survival rates for rural areas are due to a combination of inadequate resource allocation and socioeconomic disparities. These factors build on each other and each failure within the network makes the implications of the other factors more intense. While some attempts are being made to minimize the effect of each failure, eliminating these factors will require a significant upheaval in the current rural healthcare system. However, until that happens, the effects of each failure can be minimized through a series of imperfect courses of action such as those outlined in the Unger paper.

Future work should consider exploring programs that are already in place to aid rural communities with cancer treatment. Existing programs are typically good baselines for the scope of a program as most programs seek to address one concern that is present. Some of these programs include financial aid programs to assist with treatment costs for low-income patients that are either uninsured or underinsured, the role of telehealth programs, and the programs set forth in the National Rural Oncology Collaborative (NROC).

The research that I have conducted throughout this paper was done to highlight the systemic issues that result in worse cancer patient outcomes in rural areas. Noticing the effects of each underlying issues allows for a better identification and understanding of the problem. The approach to solving this issue is like combating an illness: treating the symptoms until you have learned enough to develop the cure.

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